

Meditation Detection Using Sensors from Wearable Devices

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ABSTRACT

Meditation is a practice that aims at self-inducing a state of calmed rest. In this work, we analyze physiological signals collected with wearable sensors to observe if meditation has a noticeable effect on the human body response and if this effect is inversely related to stress and can be detected using the same biosignals and similar features and methods. Our work is based on the extraction of statistical and physiological features and extends the models found in the literature by extracting 30 additional features related to heart rate variability. The results show that using wrist wearable devices, meditation periods can be distinguished from spontaneous rest with an accuracy of up to 86% accuracy.

CCS CONCEPTS

• **Computing methodologies** → *Machine learning*; • **Applied computing** → *Life and medical sciences*.

KEYWORDS

Biosignals, datasets, wearable sensors, meditation detection

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1 INTRODUCTION

Meditation consists of a set of techniques that aims at self-inducing a state of calmed rest. It has been used for relaxation and concentration purposes. Meditation has shown to have effects in the control of stressful situations, a critical skill for the short and long-term well-being of people. Several studies indicated that even short

meditation exercises may have an immediate effects by decreasing electrothermal activity [18], increasing the skin temperature, by increasing peripheral blood flow [19], or decreasing instantaneous heart rate variability (HRV) [3].

In this work, we analyse physiological signals collected using wearable sensors to observe if meditation has a noticeable effect in the human body response, and if this effect is inversely related to stress and can be detected using the same biosignals and similar features and methods [1][9]. Our main contribution aims to investigate if we can distinguish spontaneous rest affective states (baseline) from meditation, an affective state similar to rest in terms of physiological behavior, but that is being actively self-induced. For this purpose, we extend traditional feature extraction methods and enhance them by adding important physiological features, such as heart rate variability, which we believe to be more relevant than pure signal statistical features.

2 RELATED WORK

Affective state detection using physiological and physical biosignals has focused on collecting small scale data and binary classifying affect states, such as, for example stressful situations [5]. For this task, several relevant datasets have been published in recent years, e.g. (SWELL-KW [11], WESAD [15]). Most of the datasets are considered multimodal, as they contain signals from multiple sensors.

In this context, a lot of work has focused in checking different perspectives to classify stress. Those include exploring the best modalities [7][8], selecting the best features [10], finding the best sampling strategies [17], or analysing the best window sizes for the analysis of the signals [16].

Many recent studies have deepened in the idea of using commercially available wearable wrist devices to improve the continuous monitoring of people. They focus on predicting stress comparing it to resting spontaneous neutral states [16], while few studies investigated the discrimination among several affective states such as amusement. Our work aims at extending previous work by detecting periods of self-induced rest due to meditation by comparing them to spontaneous resting states.

3 EXPERIMENTAL SETUP

We perform extensive experiments and validate them in publicly available data, in this case the WESAD database [15] that contains recordings during meditation periods. The dataset contains data of 15 participants, 2 females and 13 males. The subjects are healthy, with ages that range from 24 to 35 years (mean 27.5 ± 2.4 years). The

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data is collected while performing a two-hour long study protocol composed of different tasks, representative of several baselines and three different conditions: neutral, stress, and amusement. The protocol presents two similar versions composed by the same stages, but in different order: preparation, baseline (neutral) condition, stress condition, amusement condition, meditation and recovery. The two different protocol versions and associated examples of BVP signals can be seen in Figure 1. In both versions of the study protocol, the participants were guided through meditation exercises after a rest period before the stress and amusement states. The meditation exercises had the goal to “de-excite” participants and regularize their physiological signals. The WESAD dataset focused on the discrimination of stress (and additionally amusement) from the baseline, defined as a spontaneous state of rest.

For wide applicability, our study only considers data from a wearable wrist device, the Empatica E4, which records blood volume pulse (BVP) at 64 Hz, Electro dermal activity (EDA) and skin temperature (TEMP) at 4 Hz, and three-axis acceleration (ACC) at 32 Hz sampling rate. Examples of the different signals and characteristics are shown in the Figure 2.

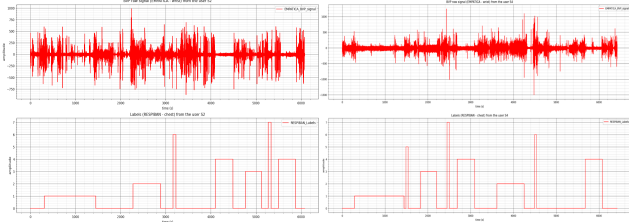


Figure 1: BVP signals from the wrist device of subjects 2 (left) and 4 (right) in WESAD. We can observe both versions of the study protocol (A and B respectively) and the labelling along the signals for the different affective states [15].

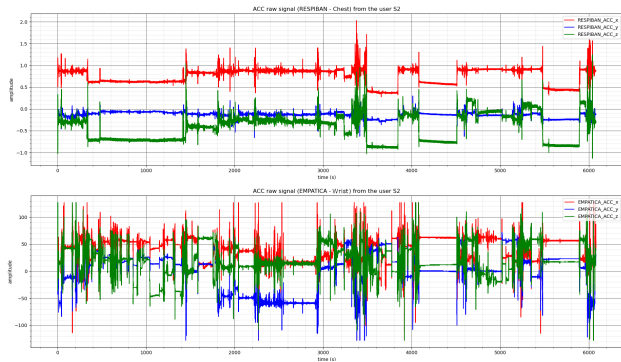


Figure 2: Differences between IMU signals from chest and wrist devices. We can observe the amount of noise and compare behaviours and correlations for the same subject S2.

4 METHODS

In this research, we perform a classification task that includes several steps: the collection and division of data in a suitable evaluation protocol, the preprocessing of the data, the training of the models, and the evaluation of the models.

4.1 Data pre-processing

Preprocessing is a critical step to prepare the data that is going to feed the classifiers that train our models. Raw data preprocessing cleans, filters, prepares, selects, and adapts the available data to extract features and patterns in the most efficient, balanced and fair way. Feature data preprocessing, which feeds directly the classifiers, fills missing and *NaN* values, and removes excessive noise or outliers. It transforms the values by normalization or standardization, filtering, and dropping undesirable data.

We follow the similar approach than the original WESAD manuscript by *Schmidt et al.* [15]. We separate the acceleration (ACC) signals in *x*, *y* and *z* components using a FIR filter to remove the high frequency noise. We also apply a low-pass filter to the raw EDA signal. Afterwards, we separate the EDA signal into different sub-components. The original approach decomposed the EDA signal in two components (namely, phasic and tonic) using the method proposed by *Choi et al.* [2]. In our case, we opted for to decompose it using the *cvxEDA* approach [6] that results in an additional component called SudoMotor Nerve Activity (SMNA), which has been shown to have classification ability for active affect states [4]. After that, we group all samples of each signal by affective state, to be able to process and extract features from the signals by affective state. The raw data preprocessing results on 9 different one-dimensional signals (*EDA*, *EDA_{phasic}*, *EDA_{tonic}*, *EDA_{smna}*, *ACC_x*, *ACC_y*, *ACC_z*, *TEMP*, *BVP*) to extract features from.

For feature data preprocessing, both *NaN* and *infinite* values were changed to reflect the values of the previous valid window. In addition, we have applied normalization based on the *RobustScaler* included in the Python package *Scikit-learn* [14], since it showed slightly better results than other approaches.

4.2 Feature extraction

To train our model, we have extracted features using different configurations related to the window length and sliding window size along each 1D signal. We have extracted features using windows of 30, 60, and 120 seconds, and a fixed sliding window of 0.25 seconds, resulting in over 140–170k windows depending on the configuration.

Following both approaches proposed by the original dataset article [15] and a study focusing on wrist signals [16], we extract 5 statistical features (min, max, mean, std, dynamic range) per signal, but we extend them by computing both the median and slope for every window. Additionally, we extracted 16 physiological features [13] and 4 statistical features derived from them (mean and std of the Heart Rate and Heart Rate Variability). We expand the set with 30 additional features. The resulting extracted features for each signal are listed in the Table 1, depicting the original features in black and the extended features in red.

4.3 Experimental protocols

We have conducted binary and 3-class classification experiments. We first provide baseline classification results between stress and non-stress, and meditation and non-meditation and affective states. Complementarily, we try to classify periods belonging to only the classes corresponding both to resting states: *meditation vs baseline*. In this case, the classifier should learn to discriminate between spontaneous rest and a self-induced state achieved through active

Table 1: List of features extracted from the signals of each sensor of the Empatica E4 device. Extended features proposed in red, originally used features in black.

Sensor	Extracted features from the sensor raw signal
Electrodermal activity (EDA)	mean, std, min, max, dyn_range, median , phasic_mean, phasic_std, phasic_min, phasic_max, phasic_dyn_range, phasic_median , tonic_mean, tonic_std, tonic_min, tonic_max, tonic_dyn_range, tonic_median , smna_mean , smna_std , smna_min , smna_max , smna_dyn_range , smna_median , slope
Acelerometer (ACC)	x_mean, x_std, x_min, x_max, x_dyn_range, x_median , y_mean, y_std, y_min, y_max, y_dyn_range, y_median , z_mean, z_std, z_min, z_max, z_dyn_range, z_median , x_peak_freq, y_peak_freq, z_peak_freq
Skin temperature (TEMP)	mean, std, min, max, dyn_range, slope, median
Blood volume pulse (BVP)	mean, std, min, max, dyn_range, median , peak_freq
Blood volume pulse (HRV)	HR_mean, HR_std, NNI_mean, sdn, NNI50, pNN50, TINN, rms, LF/HF_ratio, ULF_relative_power, LF_relative_power, HF_relative_power, UHF_relative_power, total_power, UHF_energy, LF_energy, HF_energy, UHF_energy, LF_nu, HF_nu, NNi20 , pNN20 , HR_max , HR_min , IBI , sdsd , SD1 , SD2 , SD1/SD2 , NNi_median , NNi_diff_mean , NNi_diff_max , NNi_diff_min , NNi_range , area_ellipse , cvsd , cvnni , csi , cvi , SD2²/SD1 , TRI_index , breath_rate , LF_perc , HF_perc , VLF_perc , LF_Lomb-Scargle , HF_Lomb-Scargle , LF/HF_Lomb-Scargle

guided meditation. From a physiological and psychological point of view, they should be harder to discriminate due to the proximity in the valence and arousal scale of both states. For 3-class classification experiments, we provide results for *stress vs amusement vs baseline* and *stress vs amusement vs meditation*.

4.4 Training

We have trained a set of Machine Learning classifiers selected using an exploratory strategy that tried up to 40 different classifiers, and selected 6 classifiers based on their type and preliminary performance. We selected random forest (RF) and extra trees (Extremely Randomized Trees) from ensemble learning methods, Linear Discriminant Analysis (LDA) and its variant Quadratic Discriminant Analysis (QDA) as linear classifiers and Stochastic Gradient Descent Classifier (SDGC) and eXtreme Gradient Boosting (XGBoost) as iterative methods. For validation, we follow a Leave-one-subject-out (LOSO) [12] approach and show balanced accuracies the standard deviation of the models across users, computed as the average of several runs. We compute the results for individual signals and several combinations [16].

5 RESULTS

We performed extensive experiments using three window sizes, with and without different normalization methods, using multiple

modalities and window sizes, and several classifiers. We summarize the main findings in Table 2.

Table 2: Summary of the best results for every classification modality and window size.

Type of classification	120 seconds	60 seconds	30 seconds
Binary Stress versus Non-Stress	97,06 ± 4,77 RF (EDA+BVP+HRV)	94,00 ± 20,33 RF (EDA+BVP+HRV)	92,43 ± 11,04 ETsC (EDA+ACC+ST+HRV)
Binary Meditation versus Non-Meditation	87,15 ± 11,29 RF (EDA+BVP+HR)	84,19 ± 11,04 RF (ST+HR)	82,80 ± 6,30 LDA [EDA+ACC+ST+HR]
Binary Meditation versus Baseline	86,65 ± 14,07 ETsC (EDA+ACC+ST+HR)	84,39 ± 16,98 ETsC (EDA+ACC+ST+HRV)	82,80 ± 6,30 LDA (EDA+ACC+ST+HR)
3-Classes Baseline vs Stress vs Amusement	78,53 ± 11,83 RF (EDA+ACC+BVP+HR)	76,82 ± 19,17 RF (EDA+ACC+ST+HRV)	73,95 ± 16,78 RF (EDA+ACC+ST+HR)
3-Classes Stress vs Amusement vs Meditation	79,68 ± 13,37 RF with (EDA+ACC+ST+HR)	77,14 ± 12,62 SDGC (EDA+ST+HRV)	73,41 ± 8,47 RF (EDA+ST+HR)

Detailed results are shown in Table 3. For the binary classification between stress and non-stress (baseline + amusement data), the best results correspond to signals without normalization, and without additional statistical features such as the median values. The best results reach 97,06% ± 4,77% of balanced accuracy using the signals from only two sensors (EDA+BVP+HRV). This result corresponds to Random Forests, with ExtraTrees coming very close. When we use only one sensor, the best results come from the EDA data with 89,39% ± 15,05% of balanced accuracy. The second best modality is related to HRV features extracted from the BVP signal, with 80,30% ± 20,92% of balanced accuracy. The rest of the signals show noticeably worse results.

For the classification between meditation and non-meditation (stress + amusement data), the best results correspond to signals without normalization but using the additional statistical features and an extra third EDA subcomponent (smna) with 87,15% ± 11,29% of balanced accuracy, using the signals from the EDA and the BVP sensors (EDA+BVP+HRV). In this case, the best performance comes from a Random Forest classifier. When we use only one sensor, the best results come from HRV features computed using the BVP data with 83,38% ± 12,89% of balanced accuracy.

The most interesting experiment evaluates binary classification between meditation (self-induced rest state) and baseline data (spontaneous resting state). The best results correspond, to normalized signals, using both additional statistical features and the extra third EDA subcomponent (smna) with 86,65% ± 14,07% of balanced accuracy using the signals from the EDA, ST, ACC and HRV features from the BVP signal (EDA+ACC+ST+HRV). The best performance comes from a ExtraTrees classifier. For single sensors, the best results are using the ACC data, reaching 83,78% ± 15,71% of balanced accuracy, but closely followed by the modality HRV physiological features, reaching a 81,01% ± 14,43% of balanced accuracy.

For 3-Classes classification among stress, baseline and amusement, the best results are obtained without normalization, but with

Table 3: Binary classification meditation vs baseline with Median and EDA smna. Average recognition results balanced accuracies with standard deviation using different classifiers and sensor combinations.

Sensors	RF	LDA	ETsC	QDA	SGDC	XGBoost
ACC+EDA+ST+BVP+HRV	80,94 ± 17,20	71,42 ± 22,68	85,26 ± 15,79	67,04 ± 18,15	68,24 ± 21,11	71,02 ± 17,25
EDA+ST+BVP+HRV	81,11 ± 18,68	69,19 ± 23,70	80,82 ± 17,25	59,91 ± 25,14	70,93 ± 22,81	72,46 ± 19,62
ACC+ST+BVP+HRV	83,82 ± 16,02	79,73 ± 17,09	85,63 ± 15,12	76,73 ± 18,26	78,29 ± 19,28	79,74 ± 18,14
EDA+ACC+ST+HRV	81,87 ± 17,91	74,97 ± 23,46	86,65 ± 14,07	66,36 ± 17,07	71,05 ± 21,97	69,78 ± 18,07
EDA+ACC+BVP+HRV	77,74 ± 16,44	66,74 ± 17,86	82,74 ± 15,23	67,21 ± 17,72	60,44 ± 21,43	76,52 ± 13,93
ST+BVP+HRV	84,67 ± 16,71	79,18 ± 17,93	83,65 ± 16,65	60,46 ± 27,56	82,78 ± 15,86	67,31 ± 19,31
EDA+BVP+HRV	74,53 ± 20,89	70,79 ± 21,17	76,36 ± 18,15	58,15 ± 24,28	70,71 ± 20,40	75,57 ± 15,03
EDA+ST+HRV	78,97 ± 19,80	68,72 ± 26,17	77,70 ± 17,99	57,57 ± 25,75	71,97 ± 23,38	74,29 ± 20,97
EDA+ST+BVP	73,93 ± 23,53	74,31 ± 22,24	72,78 ± 22,97	66,32 ± 20,11	73,98 ± 20,19	78,46 ± 20,60
ST+HRV	83,66 ± 16,66	75,49 ± 20,01	78,27 ± 16,88	56,08 ± 28,67	77,92 ± 17,69	71,24 ± 16,19
EDA+BVP	72,92 ± 23,62	73,26 ± 19,18	73,72 ± 22,09	62,93 ± 21,00	74,82 ± 18,87	76,59 ± 19,35
BVP+ST	81,94 ± 16,45	78,91 ± 16,37	76,44 ± 18,36	81,49 ± 14,27	77,46 ± 16,51	73,11 ± 16,95
EDA+ST	76,95 ± 19,35	71,83 ± 21,92	73,24 ± 17,33	57,04 ± 20,62	69,97 ± 20,87	72,10 ± 19,50
EDA+HRV	75,01 ± 23,66	65,88 ± 23,45	73,50 ± 19,43	56,23 ± 26,26	70,16 ± 21,63	66,53 ± 20,66
BVP+HRV	81,01 ± 15,23	79,41 ± 16,25	81,41 ± 15,51	58,16 ± 26,37	79,26 ± 16,44	74,22 ± 18,42
ACC+HRV	84,39 ± 14,56	74,72 ± 17,84	84,64 ± 15,13	73,20 ± 20,72	70,24 ± 23,25	74,09 ± 16,32
EDA	67,13 ± 24,94	67,16 ± 22,83	70,53 ± 20,39	45,19 ± 22,49	66,68 ± 21,64	65,92 ± 18,57
BVP	77,47 ± 16,71	76,02 ± 17,81	74,31 ± 17,01	71,46 ± 19,75	73,12 ± 18,19	73,71 ± 16,63
ST	67,51 ± 22,00	67,87 ± 18,75	62,96 ± 21,85	63,40 ± 15,78	64,66 ± 17,50	65,37 ± 23,26
HRV	81,01 ± 14,43	70,49 ± 20,61	75,45 ± 15,98	54,93 ± 26,17	75,17 ± 16,29	70,45 ± 19,64
ACC	80,50 ± 15,81	78,80 ± 15,87	83,78 ± 15,71	79,50 ± 18,42	81,53 ± 13,65	77,51 ± 18,56

the additional statistical features, reaching 78,53% ± 11,93% of balanced accuracy using signals from three sensors (ACC+EDA+BVP+HRV). This result is achieved with the Random Forest classifier. When we use only one sensor, the best results for the 3-Classes classification come from the EDA data with 66,31% ± 14,91% of balanced accuracy using the XGBoost classifier, followed again by the model trained with a ExtraTrees classifier and the HRV features reaching 59,29% ± 8,38% accuracy.

For 3-class stress, amusement and meditation, the best results are obtained without normalization or additional statistical features, resulting in 79,68% ± 13,37% of balanced accuracy, using the signals from all sensors (ACC+EDA+ST+HRV). When we use only one sensor, the best results come from the EDA data again, reaching a comparable 74,35% ± 16,45% of balanced accuracy using the SGD classifier. Next is the model trained with the same algorithm and the HRV features, for a 63,11% ± 15,38% of balanced accuracy.

6 DISCUSSION

According to the results of our study, we have shown meditation is a self-induced affect state that can be detected from the analysis of biosignal data collected using a commercial wrist device. In our study, the most accurate results were obtained by using 120 seconds long window size, which was the longest window size we tried. When we decrease the size of the windows, the results get slightly worse, but still usable. We can assume that the longest windows contain more consistent information about the affective state than the shortest ones. In general, classifiers based on trees worked better than other classifiers and the best results were generally achieved by using Random Forest Trees classifier (for all experiments that combined different signal modalities), while the closely related ExtraTrees classifier shows very similar results. For smaller

windows, other classification methods such as the LDA classifier show comparable relevance. The normalization of the inputs, in general does not improve consistently the results, even worsening them for some cases. However, it seems to have a small effect for particular experiments comparing meditation with rest baseline.

Detecting meditation states can be done by using the same biosignal data and machine learning methods used to predict stress, but classifying it is more difficult. This is especially true when we want to distinguish from a spontaneous, not self-induced rest state. The addition of additional physiological features (depicted as HRV features) and decompositions of the EDA signals (depicted as SNMA) has shown to play an important role in the meditation detection. This is an interesting finding that should be taken into account in future studies of meditation.

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